| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| bending Hall IC | | | | | | | | | | |

4 Result(s)

Nidec Comment

## 5 Whys's

Date of 5 why's: **4/22/2024** Update of 5 Why's:

### PROBLEM DESCRIPTION

Problem description: **lack function of hall IC**

### OCCURENCE

| Relevant factors from FTA | 1st WHY? | 1st Check | 2nd WHY? | 2nd Check | 3rd WHY? | 3rd Check | 4th WHY? | 4th Check |
|---|---|---|---|---|---|---|---|---|
| 2.1 Hall IC mounting and bending station | The Hall IC wire touch the commutator | > | The Hall IC wire is too long and not bend | > | 'The operator did not ensure that the HALL IC legs are 100% bend after mounting inside the forks. | > | lack of training for this opeartor | X |
| 2.Hall IC wire cutting | The HAll IC wire touch the commutator | > | The Hall IC wire is too long and not cut | > | 'The operator did not Make attention that the HALL IC legs are not trimed | > | lack of training for this opeartor | X |

2 Result(s)

### NON DETECTION

| Relevant factors from FTA | 1st WHY? | 1st Check | 2nd Check | 2nd WHY? | 3rd WHY? | 3rd Check | 4th WHY? | 4th Check |
|---|---|---|---|---|---|---|---|---|
| 4.Vision camera detection | The hall ic wire is too long and not cut | > | > | The leg is in the limit zone of detection | The vision area is not configured properly. | > | The risk is not analysed | X |
| 6. training level Of operator of mounting and bending Hall IC | opeartor not formed | > | > | the tabel of skills of opeartor is not respected | lack of training and multi-skilling of operators | X | | |
| 5.Indexing pallets in camera vision station | The pallets were not positioned. | > | > | There are no tools for indexing pallets. | The risk is not considered. | X | | |

3 Result(s)

Nidec Comment

## Lesson Learned Card

### LLC Type

Date of LLC: **5/28/2024**

Logistics: ☐ Management: ☐

Product: ☐ Detection: ☑

Occurence: ☑ Process: ☐

### 1. What was the problem?

What happened?

**lack function of hall IC**

Who detected it?

### 2. What is the cause?

**Occurrence**

**The opeartor didn't cut the Hall IC legs due the lack of traning**

Quality inspector at BMW Munich

Where was it detected?

BMW Munich

How was it detected?

By checking the funciton

Why is it a problem?

Missing hall signal

When did it happen?

4/2/2024 10:00:00 PM (your local time)

How many parts ?

1.00

Non detection
The detection zone is not enough to detect this defect.
The pallets were not positioned and indexed during the test operation

## 3. Description of the applied modification

### BEFORE

No respect for the standard of skills matrix

No indexing of pallets during the test by camera

Detection area on the camera vision tester is not enough.

### AFTER

Enforcement training for the Quality agent and supervisor for respect the standard of Skills table

adding tool for indexing and postionned the pallets in Camera vision tester

Adjust the vison area by upgrade the detection zonning and increase this area .

## 4. What we learned?

### ERADICATION

What is the countermeasure that was put in place to avoid recurrence ?

For occurrence
1/Enforcement training for the Quality agent and supervisor for respect the standard of Skills table
2/Implement a training room equiped : Welders, Soldering station, Products
3/Audit weekly the training program
For non-detection
4/Adjust the viison area by upgrade the detection zonning and increase this area
5/adding tool for indexing and postionned the pallets in Camera vision tester

### CHECKING

What is the method (who, what, how) for the countermeasures?

Who : AVO
What :
*/Cheking the apply of standard by Supervisor and quality agent
*/ verification the progress of the audit in station
*/ check the efficacity the tester by master parts during the production

### SIGNALING

What is the countermeasure supposed to do or make o detecting a non conformity ?

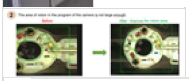
When is identifing non conformity, all transportation pallets should be
inspected

## 5. What has been modified?

- a. Proces is modified according to Before/ After sheet: ☑
- b. All documents have been updated and has been validated with the operators: ☑
- c. Procedures and work instructions: ☑
- d. Control Plan and PFMEA: ☑
- e. Maintenance instructions: ☐
- f. Records (start-up checklists, reaction sheets, preventive maintenance...): ☑
- g. Visual aids are poste
- h. Operators and other functions are traine
- i. Proof of training is avalaible (records,5 ' meetin
- j. Operators can correctly describe what they must c
- k. Oth

An evidance of modification:

Improvement action on station vison of indexing pallets .mp4



training_session (2).pdf

R_R_Vision_detection (1).pdf

Audit_station_test_vision_by_Camera (1).

CP_updtaed.pdf

PFMEA_updated.pdf

## 6. Potential applicability

LLC Applicability Product: (503-779-99-99)

LLC Applicability Line / process: CM3

LLC Applicability Plant: AVO